STA 35C: Statistical Data Science III

Lecture 16: Linear Model Selection – Regularization Methods

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Announcements

Homework 5 is posted

- Please review the problems early and ask questions as needed.
- Submit a single PDF on time and follow the submission instructions.

Office hours today

- Instructor office hours: 3:30–4:30 PM, MSB 4220.
- You are welcome to bring any questions about Homework 5 or recent lecture material.

Agenda

- **Last time:** Model selection via subset selection
  - Best subset selection: identify relevant predictors among many
  - Stepwise selection: A computationally cheaper greedy alternative
- **Today:** Regularization as an alternative to subset selection
  - Overview: what regularization is & why it can help
  - Two main examples in linear regression
    - Ridge regression
    - The lasso (least absolute shrinkage and selection operator) – today and Fri

Recap: Subset selection

Best subset selection:

- Exhaustively search all 2^p subsets
- Pick the best model for each size k, then choose among them (via R^2_{adj} or CV)
- Feasible only for smaller p (high computational cost)

Forward stepwise selection:

- Greedy approach: add one predictor at a time
- Complexity: $\mathcal{O}(p^2)$ instead of 2^p
- May miss the global optimum if local decisions are suboptimal

Backward stepwise selection:

- Greedy approach: remove one predictor at a time
- Similar pros/cons as forward stepwise

Recap: Subset selection – Comparison of search paths

An illustrative example with $p = 4$:

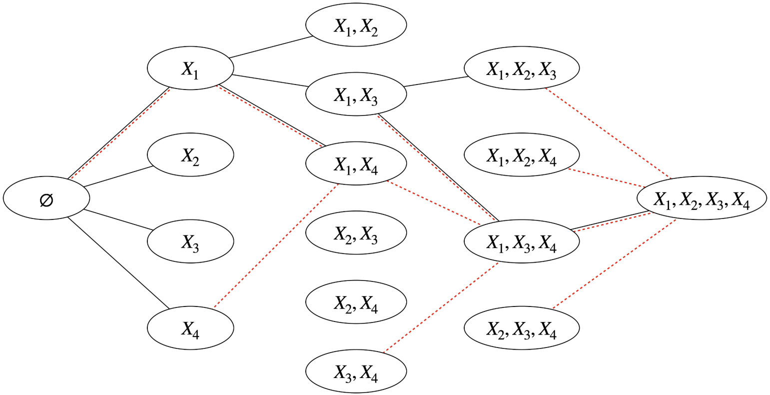

Figure: Illustration of forward stepwise (**black** solid path; successively adding $X_1 \to X_4 \to X_3 \to X_2$) and backward stepwise (**red** dashed path; removing $X_2 \to X_3 \to X_4 \to X_1$). Best subset selection checks all 2^p possibilities; all three can yield different outcomes.

Recap: Subset selection – Summary

- **Summary:**
  - **Goal**: identify a relevant subset of predictors
  - **Procedure**: evaluate subsets (all or partial), then pick best via R^2_{adj}, CV, etc.
  - BSS is exhaustive but expensive; stepwise is faster
  - Typically *refit the final chosen "best" subset using the standard least squares*
- **Advantages**:
  - Direct variable selection: some $\beta_j = 0$ (excluded)
  - Straightforward implementation and intuitive interpretation
- **Disadvantages**:
  - Even stepwise can be costly if p is very large
  - Instability: small changes in the data can alter the chosen "best" subset

$\Rightarrow$ **Regularization** controls model complexity continuously by shrinking coefficients, often giving more stable or sparser estimates without searching over all subsets

Regularization: What and why?

Recall least squares: find parameters $\hat{\beta}_0, \ldots \hat{\beta}_p$ that minimize

$$\mathrm{RSS} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \quad \text{where} \quad \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \cdots + \hat{\beta}_{ip} x_{ip}$$

- **Challenges**:
  - If p is large, predictors are highly correlated, or the data are noisy, least squares estimates can have high variance
  - This can lead to overfitting, unstable coefficients, or no unique least-squares solution when $p > n$
- **Idea**: *modify* the objective by adding a penalty on β_j's to stabilize fitting
  - Balance data fidelity *vs.* "simplicity" (by favoring smaller β_j)
  - This approach is called *regularization* (or *shrinkage*)

We will learn two prominent regularization techniques for linear regression:

- **Ridge regression** (ℓ_2 penalty)
- **Lasso** (ℓ_1 penalty)

Ridge regression: Formulation

Ridge regression: Find $\hat{\beta}_0, \ldots \hat{\beta}_p$ that minimize

$$\underbrace{\sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2}_{\text{RSS}} + \lambda \underbrace{\sum_{j=1}^{p} \beta_j^2}_{\text{penalty}}$$

- $\lambda \geq 0$ is a tuning parameter
- Each λ yields a different set of coefficient estimates $\hat{\beta}_\lambda^R$

Remarks:

- No penalty on β_0 (the intercept)
- As $\lambda \to 0$, ridge regression $\to$ standard least squares
- As λ grows, $\beta_{1:p} = (\beta_1, \ldots, \beta_p)$ shrinks toward 0
  - Reduces variance of $\beta_{1:p}$ but increases bias

Ridge regression: Illustration with 1D example

In the simplified setting with $n = p = 1$ without intercept, ridge solves for $\lambda \geq 0$:

$$\hat{\beta}^R_\lambda \in \arg\min \left\{ (y - x\beta)^2 + \lambda\beta^2 \right\}$$

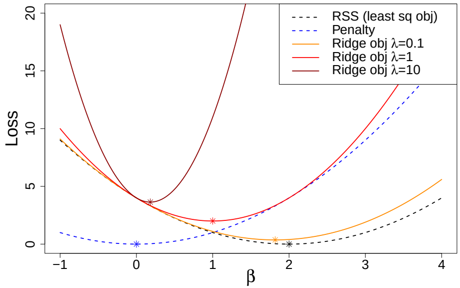

Figure: As λ grows, $\hat{\beta}^R_\lambda$ shrinks toward 0 for fixed (y, x) $(y = 2,\ x = 1)$.

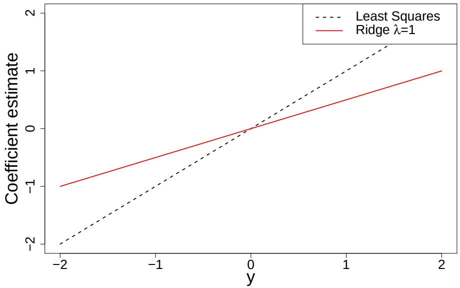

Figure: For each y, $\hat{\beta}^R_\lambda$ is smaller than the LS estimate y/x in magnitude, when $\lambda > 0$.

Ridge regression: Illustration of shrinkage effects in 2D

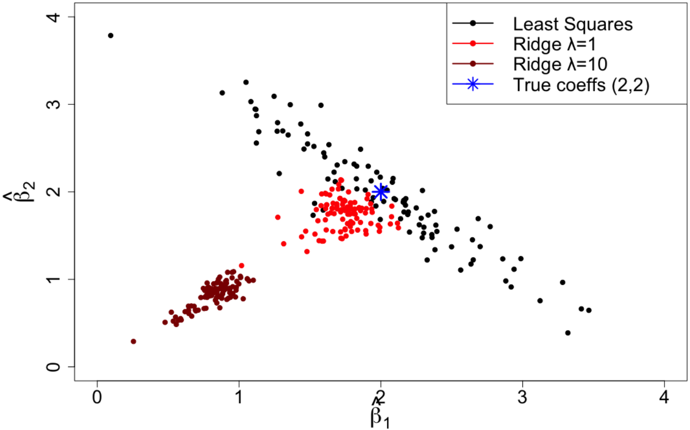

Figure: Scatter plots of 100 least squares estimates (**black**) vs. ridge estimates for $\lambda = 1$ in **red** and $\lambda = 10$ in **dark red**. As λ grows, the estimates cluster more tightly (lower variance) but shift away from the true value (blue star), indicating increased bias.

Ridge regression: Effects of scaling variables & standardization

The least squares coefficients are *scale equivariant*:

- Multiplying X_j by constant c scales $\hat{\beta}_j$ by $1/c$
- Regardless of scaling, $X_j\hat{\beta}_j$ remains the same, not affecting other coefficients

Ridge regression is *not* scale-equivariant:

- Rescaling one predictor can affect others through the penalty term
- Hence, $X_j\hat{\beta}^R_{j,\lambda}$ depends on both λ *and* predictor scaling

Therefore, we usually *standardize predictors* before ridge:

$$\tilde{x}_{ij} = \frac{x_{ij}}{s_{ij}} \qquad \text{where} \qquad s_{ij}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \bar{x}_j)^2 \quad \text{and} \quad \bar{x}_j = \frac{1}{n}\sum_{i=1}^{n} x_{ij}$$

Ridge regression: `Credit` dataset example

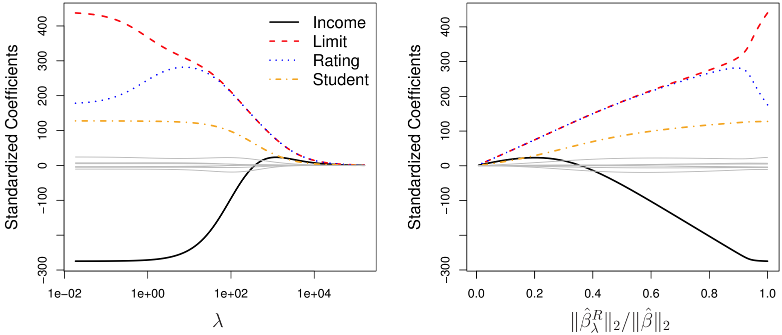

Figure: Standardized ridge coefficients for `Credit`, plotted vs. λ and $\|\hat{\beta}^R_\lambda\|_2/\|\hat{\beta}\|_2$ [JWHT21, Figure 6.4]. (Note: $\|v\|_2 = \sqrt{v_1^2 + \cdots + v_p^2}$.)

- As λ increases, all β_j shrink toward 0 (none exactly zero though)
- If variables are correlated, ridge shrinks them together in a "group" manner

Ridge regression: Advantages over least squares

Ridge's advantage rooted in **Bias-variance tradeoff**:

- $\lambda = 0$: no bias but high variance
- Larger λ: more bias but lower variance

When ridge can be most helpful:

- If least squares has high variance (e.g. $p \approx n$ or predictors are collinear)
- If data are noisy and β_j can fluctuate a lot
- For $\lambda > 0$, ridge still works even if $p > n$, producing a unique solution

Computational advantages of ridge:

- No need for 2^p model fits (as in best subset); for any λ, ridge requires only a single fit
- Indeed, we can compute solutions for all λ at near the same cost as one OLS fit

Ridge regression: Visualization of bias-variance tradeoff

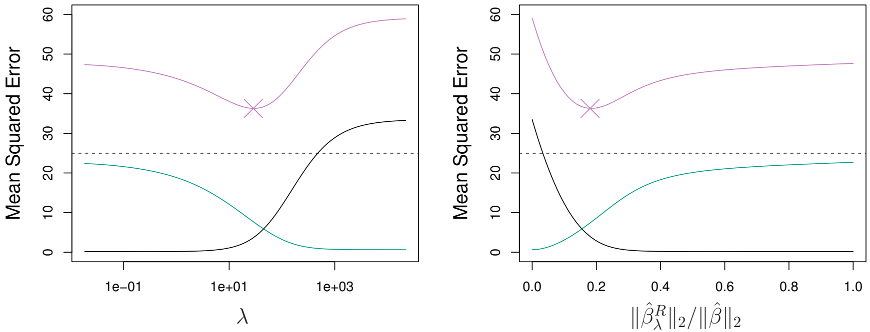

Figure: Bias-variance tradeoff in ridge for a simulated data set ($p = 45$, $n = 50$). Shown are squared bias (**black**), variance (**green**), and test MSE (**purple**) vs. λ [JWHT21, Figure 6.5].

- At $\lambda = 0$, there is no bias but high variance
- Increasing λ *significantly* reduces variance at the cost of *slightly* higher bias
- Eventually, added bias overtakes the benefit of reduced variance

Ridge regression: Selecting parameter λ

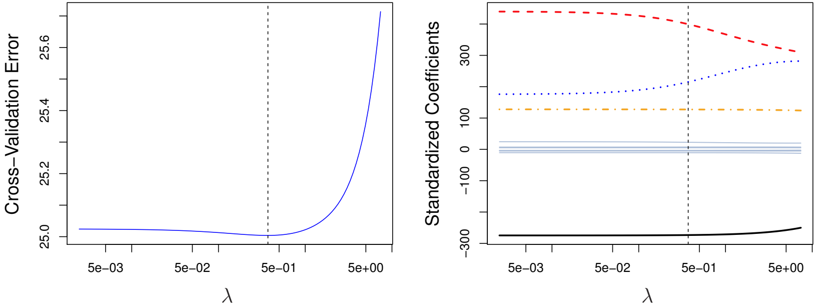

Figure: **Left:** CV errors for ridge regression on the `Credit` dataset with various values of λ. **Right:** Ridge regression coefficient estimates. The vertical dashed lines indicate the λ selected by CV [JWHT21, Figure 6.12].

- Each model with different λ is compared to each other
- Here, the chosen λ is relatively small, meaning minimal shrinkage relative to least square
- The error curve's dip is not very pronounced, suggesting a broad range of λ values yield similar performance

Ridge regression: Example with a toy dataset (R script)

- **Problem setup:**
  - $n = 5$, $p = 2$
  - Ridge vs. least squares (at $\lambda = 1$)
- **Goal:** See how ridge regression shrinks coefficients $\hat{\beta}$ compared to standard least squares

```r
# Toy data: 5 obs, 2 predictors, no intercept
df <- data.frame(
  x1 = c(1,2,3,4,5),
  x2 = c(2,1,3,1,2),
  y  = c(2,2.5,6,4,6.5)
)
```

```r
# OLS fit (no intercept => '-1')
ols_fit <- lm(y ~ x1 + x2 - 1, data = df)
cat("OLS Coeffs:\n", coef(ols_fit), "\n")

# If needed: install.packages("glmnet")
library(glmnet)

# Prepare X,y for glmnet
X <- as.matrix(df[,c("x1","x2")])
y <- df$y

# Ridge fit with lambda=1
ridge_fit <- glmnet(
  X, y, alpha=0, lambda=1,
  intercept=TRUE, standardize=TRUE
)
cat("Ridge Coeffs (lambda=1):\n", as.matrix(
     coef(ridge_fit)), "\n")
```

Pop-up quiz: Ridge regression

Which statement below is **false** about ridge regression?
(*Hint: Think carefully. Which one might not always hold?*)

A) It often lowers variance compared to least squares, at the cost of higher bias.

B) All ridge coefficients are strictly smaller in magnitude than the least squares estimates for each predictor.

C) The tuning parameter λ strongly influences ridge performance, often chosen via cross-validation.

D) Ridge tends to shrink correlated predictors together.

Answer: B.
Brief explanation:

- While ridge regression shrinks the overall magnitude of the coefficient vector, it does *not* guarantee every individual coefficient is always smaller than its least squares counterpart.
- In some cases, ridge can redistribute weights among correlated predictors, causing some estimates to exceed their OLS magnitudes even though the sum of squares $||\beta||_2^2$ is reduced.

Ridge regression: Summary

- **Ridge regression formulation**:
  - Add a penalty term $\sum_{j=1}^{p} \beta_j^2$ to the least squares objective, scaled by $\lambda \geq 0$
  - Shrinks β_j more strongly as λ grows, stabilizing estimates
- **Bias-variance tradeoff**:
  - Larger $\lambda \Rightarrow$ higher bias but lower variance
  - Especially helpful when OLS has high variance (e.g. large p, or $p > n$)
- **Computation**:
  - Efficient to solve for all λ at roughly the cost of one OLS fit
- **Limitation**:
  - Coefficients seldom reach exactly zero $\implies$ no direct variable selection

The lasso: Formulation

The lasso: Find $\hat{\beta}_0, \ldots, \hat{\beta}_p$ that minimize

$$\underbrace{\sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2}_{\text{RSS}} + \lambda \underbrace{\sum_{j=1}^{p} |\beta_j|}_{\text{penalty}}$$

- $\lambda \geq 0$ is a tuning parameter
- Each choice of λ gives a different set of Lasso estimates $\hat{\beta}_\lambda^L$

Remarks:

- No penalty on β_0 (the intercept)
- As $\lambda \to 0$, lasso $\to$ standard least squares
- As λ grows, many β_j shrink toward *zero*, and some exactly become 0

Unlike ridge, the lasso can yield exact zero estimates $\implies$ **variable selection**!

Lasso: Illustration with 1D example

In the setting with $n = p = 1$ without intercept, ridge solves for $\lambda \geq 0$:

$$\hat{\beta}^R_\lambda \in \arg\min \left\{ (y - x\beta)^2 + \lambda|\beta| \right\}$$

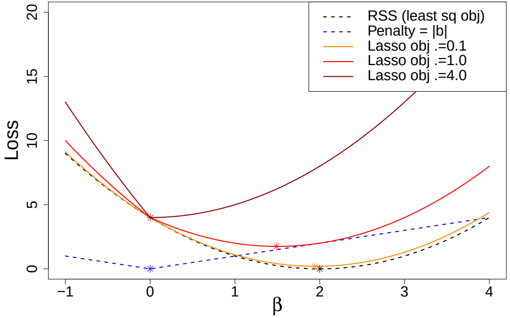

Figure: As λ grows, $\hat{\beta}^\ell_\lambda$ shrinks more aggressively; small $|y|$ can yield $\beta = 0$ ($y = 2$, $x = 1$ fixed).

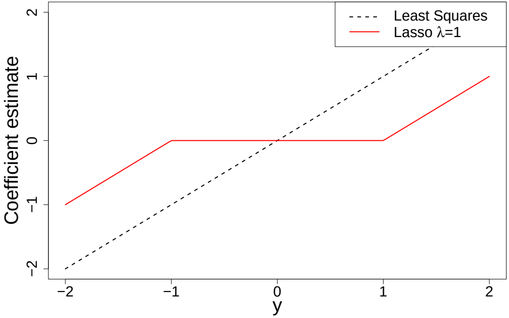

Figure: At $\lambda = 1$, $x = 1$, β hits 0 iff $|y| \leq 1$. This “thresholding” property underlies variable selection.

Lasso: Regularization reduces variance, but...

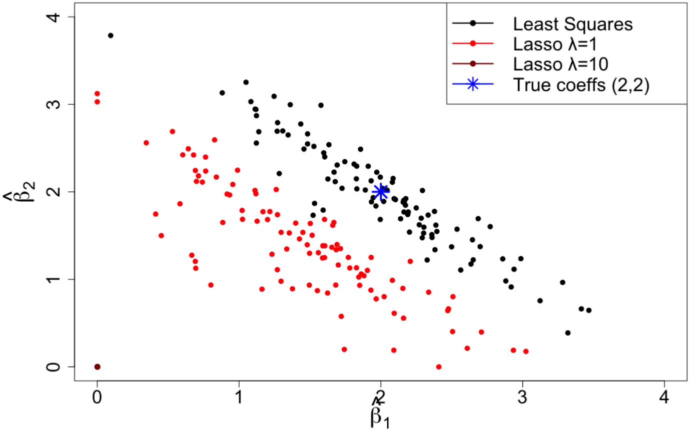

Figure: Scatter plots of 100 least squares estimates (**black**) vs. lasso estimates for $\lambda = 1$ in **red** and $\lambda = 10$ in **dark red**. Lasso can aggressively shrink or zero-out coefficients, but the variance reduction is less uniform than ridge. The shift from the true (blue star) may or may not be worth it.

Lasso: Regularization enables variable selection

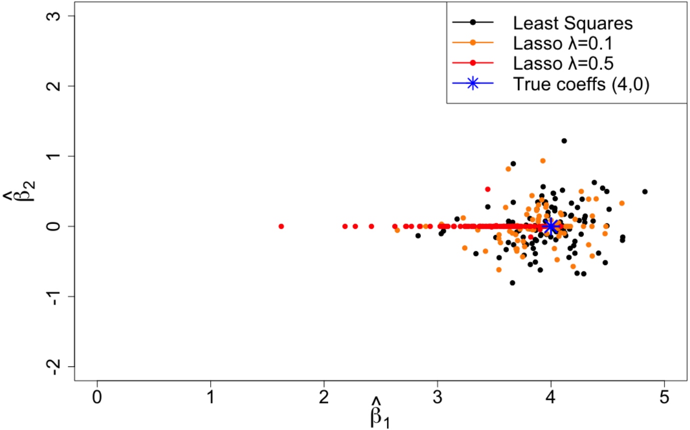

Figure: Scatter plots of 100 least squares estimates (**black**) vs. lasso estimates for $\lambda = 0.1$ in orange and $\lambda = 0.5$ in **red**. If the true $\beta_2 = 0$ (blue star), lasso can correctly select the significant variable (X_1), while suppressing noise and driving estimates to zero for X_2, thereby capturing the "sparse" true associations.

Wrap-up & Takeaways

Regularization:

- **Ridge**: ℓ_2 penalty shrinks all coefficients toward zero, good if many have modest nonzero effects
- **The lasso**: ℓ_1 penalty can set some coefficients exactly to zero (variable selection)
- **Tuning** λ typically via cross-validation for both

	Ridge	**Lasso**
Penalty	$\sum_{j=1}^{p} \beta_j^2$	$\sum_{j=1}^{p} \lvert\beta_j\rvert$
Effect	shrinks coefficients	shrinks and can set to zero
Variable selection	No	Yes
Often useful when	many small/moderate effects	only some predictors matter
Tuning	λ typically chosen by cross-validation	

Takeaway: Ridge stabilizes estimates; lasso can produce sparse, interpretable models.

References

Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani.
An Introduction to Statistical Learning: with Applications in R, volume 112 of *Springer Texts in Statistics*.
Springer, New York, NY, 2nd edition, 2021.